

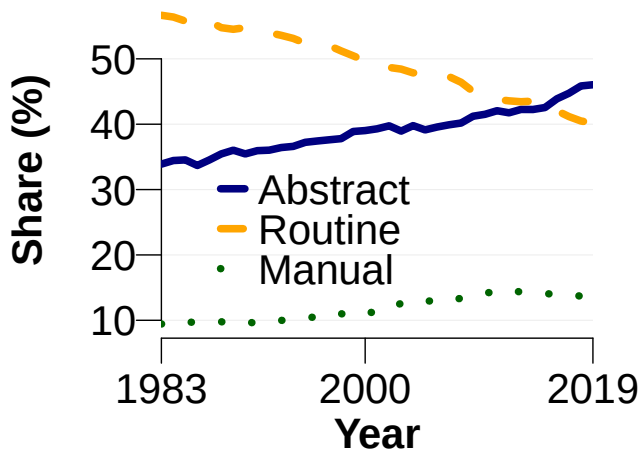
Taxing Automation?

A Case for Redistribution across Skills and Generations

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October 2, 2026, Claremont McKenna College

Labor Market Polarization



- Abstract:
 - managers, professionals
- Routine:
 - clerical, production
- Manual:
 - personal, food services

Adjustment over the Life-Cycle

- **Age and human capital affect occupational adjustment**
 - Switching destroys accumulated occupation-specific human capital
 - Older workers have fewer years to recover the cost of switching

Adjustment over the Life-Cycle

- **Age and human capital affect occupational adjustment**
 - Switching destroys accumulated occupation-specific human capital
 - Older workers have fewer years to recover the cost of switching
- **Assets and public insurance affect consumption**
 - Savings cushion earnings losses; borrowing constraints limit adjustment
 - Capital returns matter for older asset holders
- Welfare effects differ across skills and generations

This Paper

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 - Those alive in 1980 lose; the young low-skilled lose most
2. How do effects vary across generations?
 - Later high-skilled entrants gain; low-skilled losses grow
3. Can an automation tax redistribute these gains and losses?
 - Optimal equipment-income tax: **42%**, versus 10% in the baseline
 - Social welfare gain: 0.053% CEV

1. Routine technical change and polarization

- Autor et al. (2003); Autor and Dorn (2013), Bárány and Siegel (2018); Acemoglu and Restrepo (2018)
- **This paper: welfare across skills and generations**
 - ★ Workers adjust occupations, participation, consumption, and saving
 - ★ Track gains and losses over the life-cycle and for later entrants

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- **This paper: explicit aging and individual adjustment**
 - ★ Occupational human capital, individual saving, and earnings risk
 - ★ Remaining working lives and retirement affect welfare

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 - ★ Relative to GRT2022: mobility during working lives and earnings risk
 - ★ Relative to BZ2025 explicit aging, human capital, and retirement

DATA

Samples and Definitions

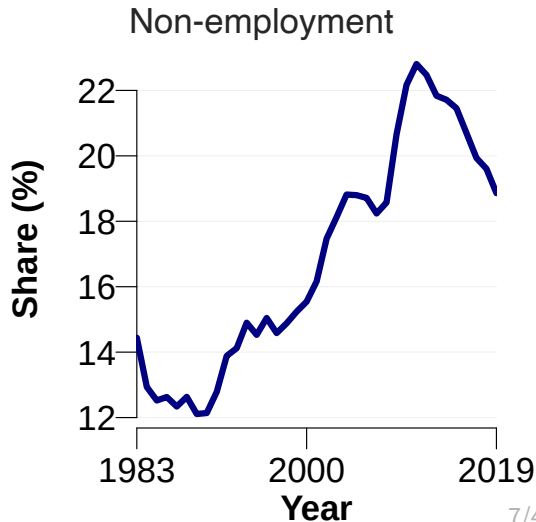
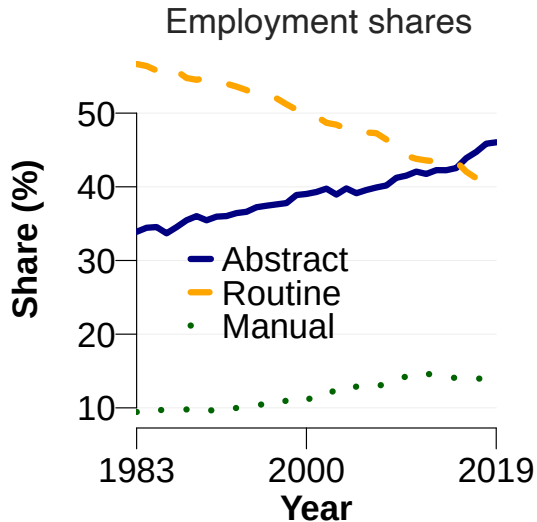
- **CPS: household heads, male and female**

- Employment trends: ages 30–59, 1983–2019
- Age profiles: ages 25–79, pooled 1983–1985; Non-emp.: non-employment
- Wages: ages 25–64, 1983–2018; full-year workers

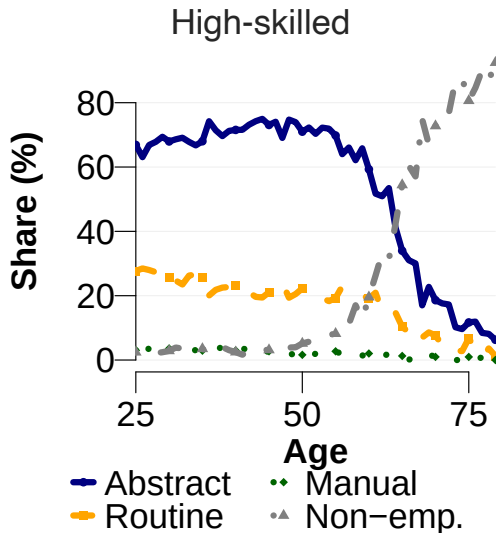
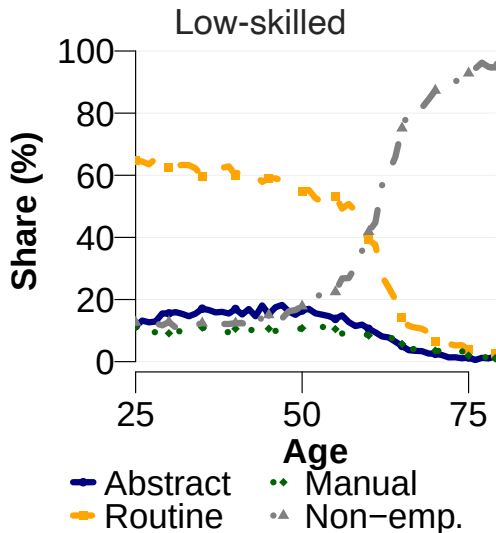
- **PSID: productivity shocks**

- Core sample, household heads aged 30–59, 1983–1996
- Low-skilled: no college degree; high-skilled: college degree
- Unemployed up to 13 weeks with a known prior occupation: employed

Employment and Non-Employment



Occupations by Age and Skill



Occupational Mobility Calibration

Low-skilled			
	A	R	M
A	96.79	2.72	0.50
R	0.78	98.68	0.54
M	0.69	3.66	95.66

High-skilled			
	A	R	M
A	98.60	1.25	0.15
R	3.53	96.09	0.38
M	4.85	3.98	91.18

- Rows: origin; columns: destination; probabilities in percent
- Six directional mobility costs for each skill type

MODEL

Model Overview

- Overlapping generations with heterogeneous workers
 - Choose consumption, saving, occupations, and participation
- Occupation-specific human capital and idiosyncratic earnings risk
- Firms combine abstract labor, routine tasks, manual labor, and capital
 - Equipment competes with routine labor in every task
- Equilibrium wages, capital returns, and task assignment

Demographics and Household States

- Annual periods; enter at age 25, live up to age 85
 - Survival risk; no bequest motive; entrants have zero assets
- Fixed education type $s \in \{L, H\}$
- State before current shocks: $x = (j, a, s, o, h)$
 - Age, assets, skill, prior occupation, human capital
- Entrants choose abstract, routine, or manual work
 - Later workers may switch or leave; non-employment is absorbing

Consumption and Saving

$$c + a' = \underbrace{(1 + r_t^H)a}_{\text{Assets and return}} + \underbrace{(1 - \tau_{l,t})y_t(d)}_{\text{After-tax earnings}} \\ + \underbrace{ss \mathbf{1}\{j \geq j_{ss}\} + CF_t(d)}_{\text{Benefits}} + \underbrace{T_t}_{\text{Transfer}}.$$

- c : consumption; a : beginning assets; $a' \geq 0$: next-period assets
- r_t^H : after-tax asset return; T_t : per-person lump-sum transfer

Earnings and Benefits

- $y_t(d) = w_{d,t}h_d\eta_{s,d}$: earnings in destination d
- $w_{d,t}$: wage per effective unit; h_d : human capital in destination d
- $\eta_{s,d}$: earnings-productivity shock; earnings are zero in non-employment
- $\tau_{l,t}$: labor-income tax; ss: benefit from age $j_{ss} = 66$
- $CF_t(d)$: means-tested payment enforcing the consumption floor

Capital Income before Tax

$$\begin{aligned}\Pi_{E,t} &= \underbrace{Y_{K_E,t}K_{E,t}}_{\text{Rental income}} + \underbrace{(1 - \delta_E)p_{E,t}K_{E,t}}_{\text{Remaining value}} - \underbrace{p_{E,t-1}K_{E,t}}_{\text{Initial value}} . \\ \Pi_{O,t} &= \underbrace{Y_{K_O,t}K_{O,t}}_{\text{Rental income}} - \underbrace{\delta_O K_{O,t}}_{\text{Depreciation}} .\end{aligned}$$

- Π_E, Π_O : aggregate pre-tax net income, in final-good units
- Y_K : marginal products; K : stocks; $\delta_E = \delta_O = 0.10$
- p_E : equipment price; other-capital price is one

Capital Taxes and Household Returns

$$r_t^H \mathcal{A}_t = \underbrace{(1 - \tau_{E,t}) \Pi_{E,t}}_{\text{After-tax equipment income}} + \underbrace{(1 - \tau_O) \Pi_{O,t}}_{\text{After-tax other-capital income}} .$$
$$\mathcal{A}_t = \underbrace{p_{E,t-1} K_{E,t}}_{\text{Equipment value}} + \underbrace{K_{O,t}}_{\text{Other-capital value}} .$$

- $\mathcal{A}_t$: aggregate beginning assets; r_t^H : a return rate
- Π_E, Π_O and $r_t^H \mathcal{A}_t$: income amounts, not rates
- Baseline $\tau_E = 10\%$, $\tau_O = 8\%$: Acemoglu et al. (2020); deducted once

Household Value Function

$$W_t(x, d, \eta_s) = \max_{c, a' \geq 0} \left\{ \underbrace{u(c)}_{\text{Current utility}} + \underbrace{\beta \phi_{j+1} V_{t+1}(j+1, a', s, d, h_d)}_{\text{Continuation value}} \right\}.$$

- $u(c) = c^{1-\gamma}/(1-\gamma), \gamma = 2; \beta = 0.9820$
- ϕ_{j+1} : probability of surviving from age j to $j+1$
- W_t : value conditional on destination d and current shocks
- V_{t+1} : value before next-period shocks and occupational choice

Occupational Choice Scores

$$z_t(x, d, \eta_s) = W_t(x, d, \eta_s) - \underbrace{(B_s + \kappa_{s,d})\mathbf{1}\{d \neq N\}}_{\text{Work costs}} - \underbrace{c_{od}^s}_{\text{Mobility cost}} .$$

- z_t : utility score; N : non-employment; o : prior occupation
- B_s : participation cost; $\kappa_{s,d}$: destination-specific cost
- c_{od}^s : cost of moving from o to d ; costs are in utility units
- θ_s : inverse preference-shock scale; ▸ Logit probabilities

Occupational Choice and Human Capital

- Staying preserves accumulated human capital and adds experience

Occupational Choice and Human Capital

- Staying preserves accumulated human capital and adds experience
- Switching resets human capital to the destination's entry level
 - Or its age-specific level, if lower
- Occupational choice reflects wages, continuation values, and costs
 - Participation costs, occupation-specific costs, directional mobility costs
 - Productivity shocks and preference shocks
- Young workers have a longer horizon over which to benefit from switching

Production

$$Y_t = K_{O,t}^\alpha X_t^{1-\alpha}$$

$$X_t = \left[A_{A,t} L_{A,t}^{\frac{\sigma-1}{\sigma}} + A_{R,t} Z_t^{\frac{\sigma-1}{\sigma}} + A_{M,t} L_{M,t}^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}$$

- Y : final output; K_O : other capital; X : labor/task composite
- L_A, L_M : abstract/manual labor; Z : routine tasks; $A_{o,t}$: production coefficients
- $\alpha = 0.2648$: other-capital share; $\sigma = 0.60$: substitution elasticity

Routine-Task Production

$$Z_t = \left[\int_0^1 q_t(i)^{\frac{\zeta-1}{\zeta}} di \right]^{\frac{\zeta}{\zeta-1}}, \quad \zeta = 0.50$$

$$q_t(i) = k_t(i) + A_{\text{task}} i^{\chi} \ell_t(i), \quad \chi = 2.12$$

- ζ : task elasticity; χ : comparative advantage (Acemoglu et al., 2020)
- $k_t(i), \ell_t(i)$: equipment, routine labor; totals $K_{E,t}, L_{R,t}$
- $q_t(i)$: task output; A_{task} : scale; labor productivity rises in i

Endogenous Task Assignment

$$\underbrace{u_{E,t}}_{\text{Equipment unit cost}} \quad \text{vs.} \quad \underbrace{\frac{w_{R,t}}{A_{\text{task}} i^\chi}}_{\text{Labor unit cost}}$$

$$\iota_t = \left(\frac{w_{R,t}}{A_{\text{task}} u_{E,t}} \right)^{1/\chi}, \quad 0 < \iota_t < 1.$$

- w_R : routine wage; u_E : equipment rental cost; ι_t : automated share
- Equipment performs $i < \iota_t$, labor $i > \iota_t$; a higher u_E lowers ι_t

Rental Costs and the Common Return

$$u_{E,t} = \left(1 + \frac{r_t}{1 - \tau_{E,t}}\right) p_{E,t-1} - (1 - \delta_E) p_{E,t}, \quad u_{O,t} = \frac{r_t}{1 - \tau_O} + \delta_O.$$

- u_E, u_O : rental costs; r_t : common after-tax return rate
- When capital adjusts, positive stocks satisfy $Y_{K_E,t} = u_{E,t}$ and $Y_{K_O,t} = u_{O,t}$

$$(1 - \tau_{E,t}) \Pi_{E,t} = r_t p_{E,t-1} K_{E,t},$$

$$(1 - \tau_O) \Pi_{O,t} = r_t K_{O,t} \implies r_t^H = r_t.$$

- At the surprise shock, fixed inherited stocks earn market-clearing rents

Government Budget

$$\underbrace{G_t + H_t + N_t T_t}_{\text{Outlays}} = \underbrace{\tau_{l,t} Y_{L,t}}_{\text{Labor tax}} + \underbrace{\tau_{E,t} \Pi_{E,t} + \tau_O \Pi_{O,t}}_{\text{Capital taxes}} + \underbrace{\varepsilon_t}_{\text{Bequests}} .$$

- G_t : government purchases; H_t : total social security and floor payments
- N_t : population; T_t : per-person transfer; $Y_{L,t}$: total labor earnings
- ε_t : accidental bequests collected by the government
- Baseline $\tau_l = 25\%$: Acemoglu et al. (2020); no government debt

Market Clearing

$$\underbrace{Y_t}_{\text{Output}} = \underbrace{C_t}_{\text{Consumption}} + \underbrace{p_{E,t}l_{E,t} + l_{O,t}}_{\text{Investment spending}} + \underbrace{G_t}_{\text{Purchases}} .$$

- C_t : aggregate consumption; $l_{E,t}, l_{O,t}$: capital investment quantities
- $K_{b,t+1} = (1 - \delta_b)K_{b,t} + l_{b,t}, b \in \{E, O\}$
- Asset market: household assets equal capital value $\mathcal{A}_t$
- Effective labor supplied in each occupation equals firms' demand

CALIBRATION

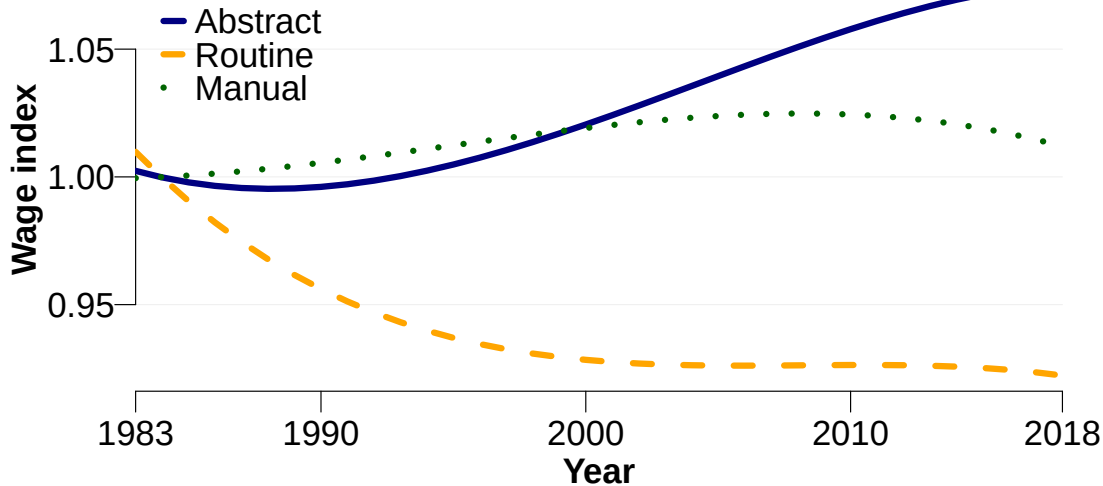
Calibration Strategy

- **Initial economy: early 1980s** ▸ Parameters ▸ Mobility
 - Entry shares, mobility, participation at age 65; wealth/output = 2.7
- **Household inputs**
 - CPS human capital profiles; PSID productivity shocks
- **Technology along the transition**
 - Recover occupational productivity to match estimated wages
 - Equipment value/output identifies the constant task scale
 - Wage index: 1983–1985 = 1; production indices: 1980 = 1

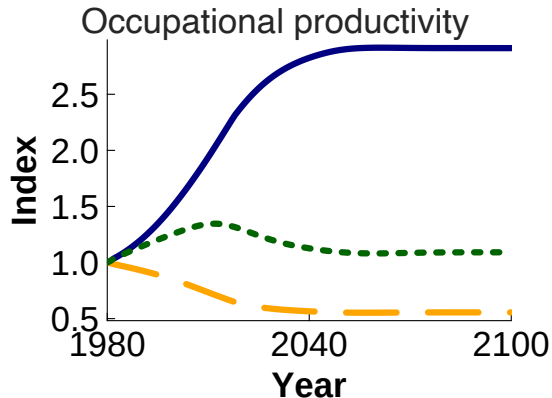
Calibration Strategy

- **Reading the initial-fit figures**
 - Solid lines: model; dashed lines: CPS data
- **Limitation: midlife non-employment**
 - Low-skilled at age 45: 1.4% in the model; 15.1% in CPS

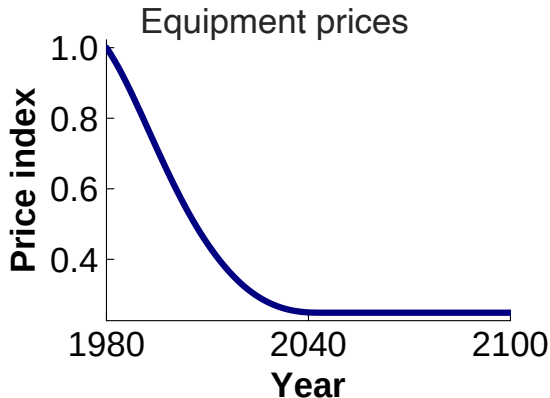
Estimated Occupational Wages



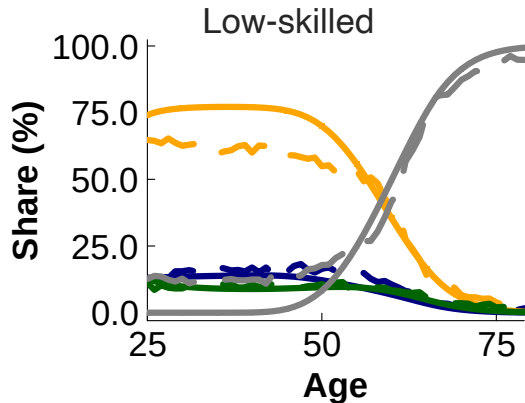
Occupational Productivity and Equipment Prices



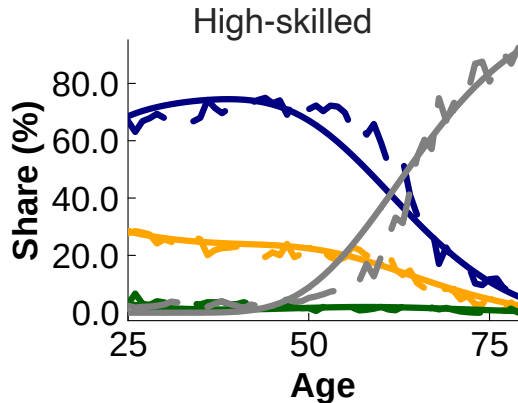
— Abstract - Manual
— Routine



Initial Economy: Occupational Shares



— Abstract — Manual
— Routine — Non-emp.



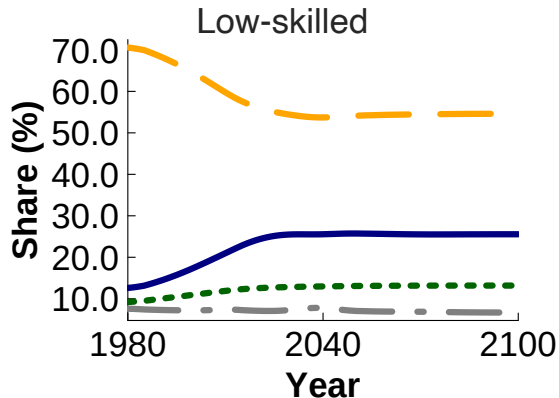
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TRANSITION AND WELFARE

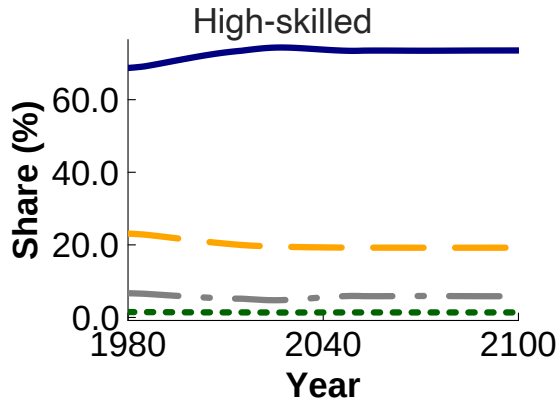
Transition and Welfare Comparison

- Common initial allocation in 1980; new paths learned in 1981
- Baseline: changing technology and equipment prices
 - Comparison: both fixed at 1980 values
- Both: same demographics, preferences, and tax rates
 - Wages and returns clear markets; government spending balances budgets
- Welfare: consumption equivalent variation (CEV)
 - Positive CEV: gain in the baseline; includes the 1980 utility flow
 - All: skills pooled; H/L: high-/low-skilled

Occupational Transition

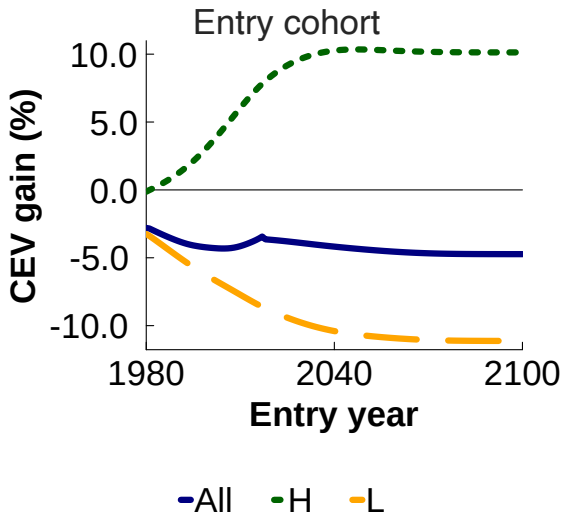
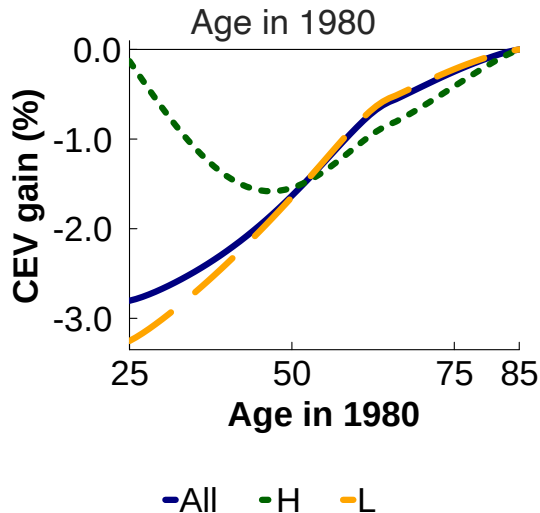


— Abstract - Manual
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Welfare Effects of Automation



AUTOMATION TAX

Equipment-Income Tax Experiment

- Permanent tax τ_E on **net equipment income**, starting in 1981
- Baseline tax rates: Acemoglu et al. (2020)
 - Equipment: 10%; labor: 25%; other capital: 8%
- Return revenue through uniform lump-sum transfers T_t
 - Labor and other-capital taxes fixed; government spending on baseline path
- Same technology, equipment prices, demographics, and benefit rules

Planner's Problem

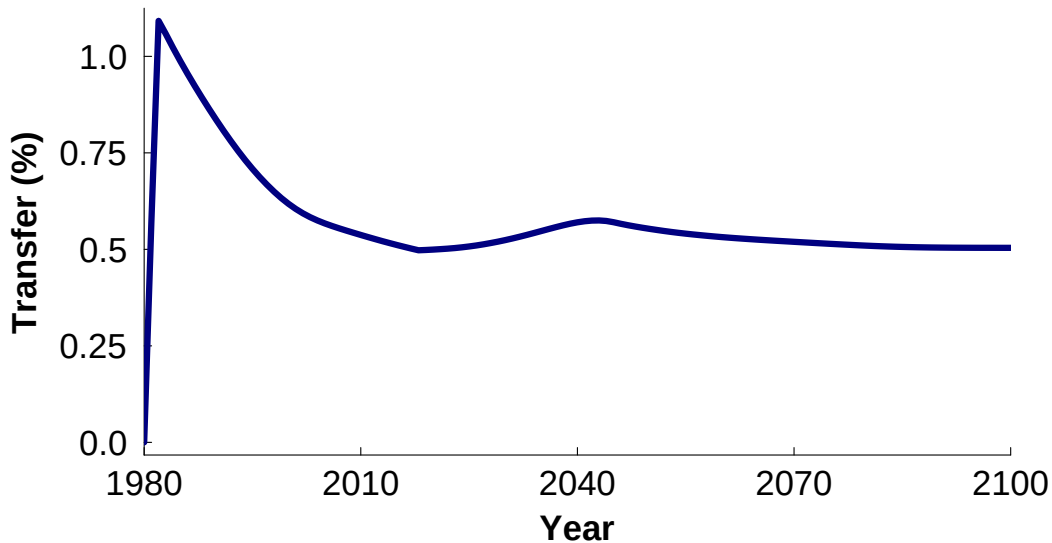
$$\max_{\tau_E} \mathcal{W}(\tau_E) = (1 - \beta) \left[\underbrace{\sum_{j,s} N_{j,s,0} U_{j,s,0}^{\text{old}}(\tau_E)}_{\text{Current generations}} + \underbrace{\sum_{t \geq 1} \beta^t \sum_s N_{s,t}^{\text{ent}} U_{s,t}^{\text{ent}}(\tau_E)}_{\text{Future generations}} \right]$$

- $N_{j,s,0}$: 1980 mass by age j and skill s , including 1980 entrants
- $U_{j,s,0}^{\text{old}}$: remaining expected utility from 1981, discounted to 1980
- $N_{s,t}^{\text{ent}}$: entrants in 1980 + t ; β^t discounts to 1980 ▸ Stationary tail
- $U_{s,t}^{\text{ent}}$: expected lifetime utility at entry; both include survival

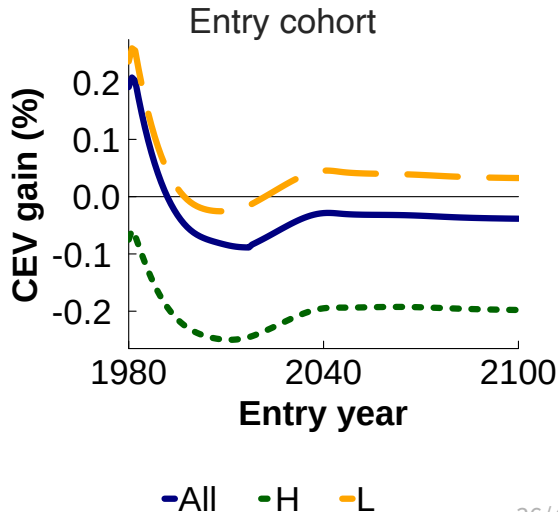
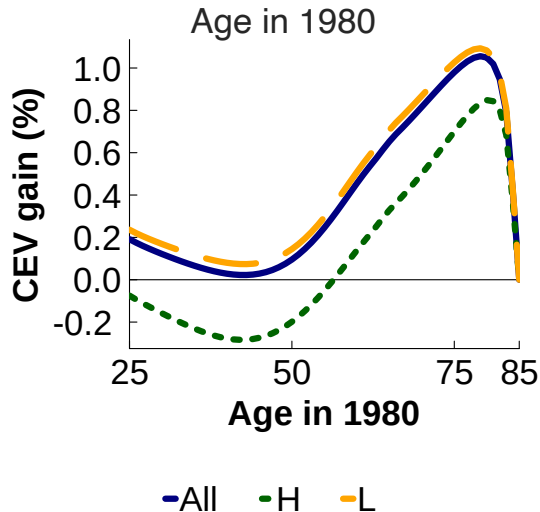
Optimal Equipment-Income Tax

- Optimal equipment-income tax: **42%**, versus 10% baseline
- Social welfare gain: **0.053% CEV**
 - U : consumption, work and mobility costs, preference shocks
 - Same consumption change for everyone in $\mathcal{W}$, from 1981
 - CEV rises with $\mathcal{W}$: the same τ_E maximizes both
- Entrants in 1980: **0.19% CEV**
 - Low-skilled: 0.24%; high-skilled: -0.07%
 - Cohort CEV includes the common 1980 flow; $\mathcal{W}$ omits it

Lump-Sum Transfers (% of Average Consumption)



Optimal Tax: Welfare Effects



Optimal Tax: CEV Decomposition

$$U = C + D + T$$

- C : consumption utility; D : minus work and occupation costs
- T : taste utility; equipment-income tax: 42% versus 10%

	Total CEV (%)	Consumption	Costs	Taste
Social welfare	+0.0531	-2.7	143.5	-40.8
High-skilled, all generations	-0.1821	87.0	40.0	-27.0
Low-skilled, all generations	+0.1216	36.4	98.3	-34.8
High-skilled, alive in 1980	-0.1040	81.4	15.2	3.4
Low-skilled, alive in 1980	+0.2572	62.6	52.6	-15.2

- Components: share of total CEV (%); sum to 100%, up to rounding
- Negative shares offset the total change; utility from 1981

Equipment Flow Tax Experiment

$$\tau_{FP_{E,t}}[K_{E,t+1} - (1 - \delta_E)K_{E,t}] = \tau_{FP_{E,t}}I_{E,t}.$$

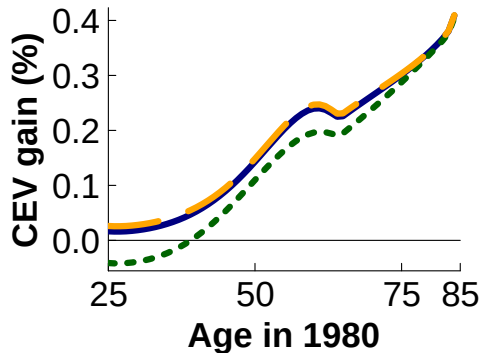
- Permanent tax on equipment investment, including replacement, from 1981
- Uniform lump-sum transfers return revenue; refund negative investment
- Fixed $(\tau_E, \tau_O, \tau_I) = (10, 8, 25)\%$: Acemoglu et al. (2020)

	Social CEV	Low-skilled social CEV
Social optimum: 2%	+0.003%	+0.018%
Low-skilled optimum: 17%	-0.113%	+0.003%

- CEV from 1981: consumption utility, work and mobility costs, taste utility

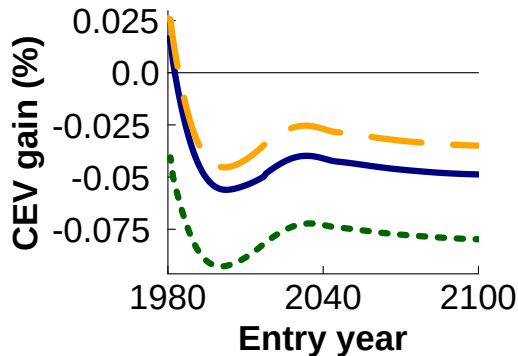
Equipment Flow Tax: Welfare at 2%

Age in 1980



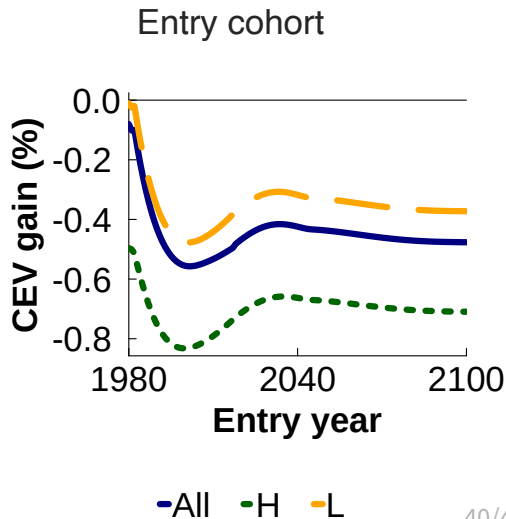
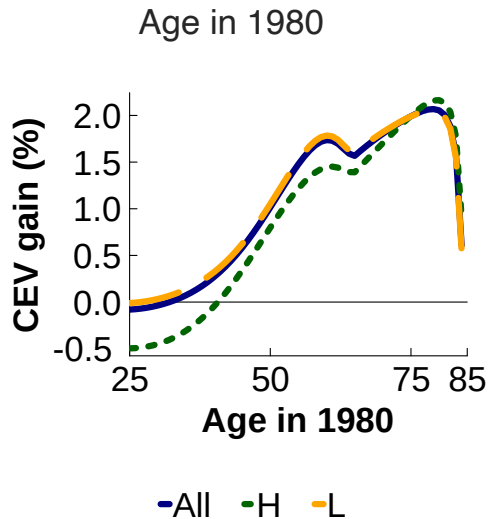
- All - H - L

Entry cohort



- All - H - L

Equipment Flow Tax: Welfare at 17%



CONCLUSION

Conclusion

- **Automation redistributes welfare across skills and generations**
 - Later high-skilled entrants gain; low-skilled entrants lose
 - Both skill groups entering in 1980 lose from automation
- **Equipment taxation redistributes automation gains**
 - Optimal tax: 42%; social welfare rises by 0.053% CEV
 - Low-skilled entrants in 1980 and 2100 gain; high-skilled lose

APPENDIX

Calibration Parameters

Parameter	Value	Target or assumption
β	0.9820	Wealth/output: 2.7
B_L, B_H	1.69, 0.56	Participation at age 65
θ_L, θ_H	0.22, 0.06	Routine entry-share change
σ, ζ	0.60, 0.50	Substitution elasticities
δ_E, δ_O	0.10, 0.10	Depreciation
χ	2.12	Task comparative advantage
α	0.2648	Other-capital share
$\underline{c}$	0.10	Initial average earnings: 1
ss	0.38	Benefit from age 66

Occupational Choice Probabilities

$$\Pr_t(d \mid x, \boldsymbol{\eta}_s) = \frac{e^{\theta_s z_t(x, d, \boldsymbol{\eta}_s)}}{\sum_{k \in \mathcal{D}(o)} e^{\theta_s z_t(x, k, \boldsymbol{\eta}_s)}}$$

$$V_t(x) = \int \frac{1}{\theta_s} \log \left[\sum_{d \in \mathcal{D}(o)} e^{\theta_s z_t(x, d, \boldsymbol{\eta}_s)} \right] dF_s(\boldsymbol{\eta}_s)$$

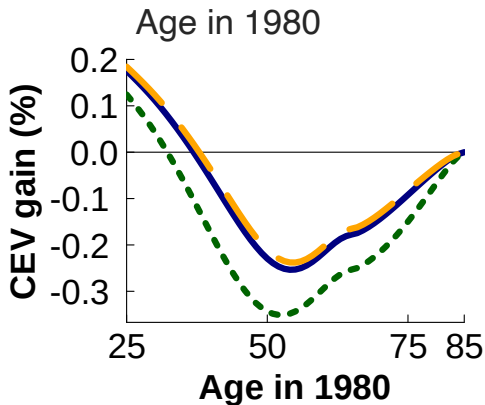
- $\mathcal{D}(o)$: available destinations; F_s : joint earnings-shock distribution
- θ_s : inverse preference-shock scale; z_t : occupational utility score

Stationary-Tail Welfare Calculation

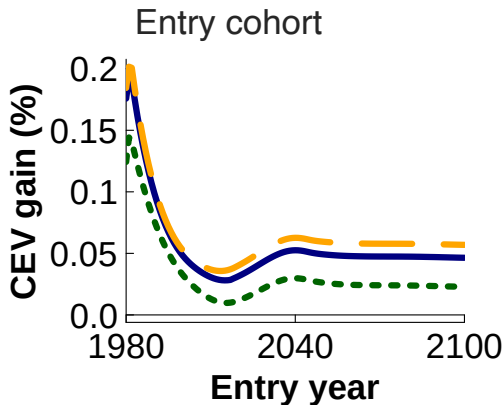
$$\sum_{t=\bar{t}+1}^{\infty} \beta^t \sum_s N_{s,*}^{\text{ent}} U_{s,*}^{\text{ent}} = \frac{\beta^{\bar{t}+1}}{1-\beta} \sum_s N_{s,*}^{\text{ent}} U_{s,*}^{\text{ent}}.$$

- $\bar{t}$: last explicitly evaluated entry year, measured from 1980
- $N_{s,*}^{\text{ent}}, U_{s,*}^{\text{ent}}$: stationary entrant masses and lifetime utilities
- Objective: $\bar{t} = 220$ (2200); check: $\bar{t} = 260$ (2240)
- Apply the same weights to consumption and nonconsumption utility

Welfare with Labor-Tax Adjustment



— All - - H — L



— All - - H — L